Topic 1: Gram-Schmidt Decomposition

Question 1:

Given the vectors
v1 = [1, 1]
v2 = [1, 0]

Write a Python program using NumPy to perform the Gram-Schmidt process.

1. Convert the vectors into NumPy arrays.
2. Find the orthogonal vectors u1 and u2.
3. Normalize the vectors to obtain an orthonormal set q1,q2.
4. Print the resulting orthonormal vectors.

Question 2:

Given the vectors
v1 = [1, 1, 0]
v2 = [1, 0, 1]
v3 = [0, 1, 1]

Write a Python program to perform Gram-Schmidt decomposition.

1. Compute the orthogonal vectors u1,u2,u3.
2. Normalize them to obtain q1,q2,q3.
3. Construct the matrix Q using the orthonormal vectors as columns.

Question 3:

Given the matrix

A = [[1, 1, 1],
[1, 2, 3]
[1, 3, 6]]

Write a Python program to find the QR decomposition of A using the Gram-Schmidt process.

1. Extract the columns of A.
2. Apply Gram-Schmidt to obtain an orthonormal basis Q.
3. Calculate R = Q.T @ A